

Meeting Notes for Feb 9 and Feb 11

Date: February 9 and February 11

Time: 1:30 - 3:00 PM on February 9 and February 11 (PDT)

Attendees: Prof. Jan, Dr. Javed, Dr. Yue and Feng

Topic: In-person meeting to discuss the work done so far and next steps toward completing the thesis study; Feng's mid-term report of the thesis study.

Summary of Discussion

0.1 Meeting on Feb 9

- The first in-person meeting, Prof. Jan showed the background and the overall framework of Jian's study. Released by point sources, the toxic chemicals that across the SCDRS and the DR area and their poisonous effect on the creatures constitute the ecological focus of the study.
- The **5 clusters** from two-case studies are the specifically environmental homogeneous groups of taxa, in each of which the Zoobenthic Community Indicator (ZCI) was designed and developed respectively from its taxa matrix.
- The dissimilarity of taxa composition between sites **cannot be derived** well from the original measures. The uncontrollable factors in sampling process, the creatures' migrating behavior and so on, can lead the recorded numerical values and the following quantitative operations in **Euclidean Geometry view**, not reflect the real dissimilarity of taxa composition between sites.

A typical example is that when two sites **both have 0** abundance of a certain taxa, this **artificially** inflates similarity under the original metric. But this 'double-zero' **does not** have to mean the two sites are same or similar in conditions for holding that taxa, such absences may reflect temporal variation, sampling artifacts, or species behavior rather than shared environmental conditions.

For these reasons, transformations on taxa matrix are necessary and **warrant careful treatment**. Such transformations engage in several steps in the whole computational process, including a) clustering on taxa matrix, b) calculating the ZCI values.

- Jian replied with email and showed interests in talking about the Feng's thesis work, the meeting time with Jian is pending to be decided, but it would not be too late over this month.

0.2 Meeting on Feb 11

- Feng delivered a mid-term progress report on his thesis research, it took about 40 minutes and was presented in the structured as follows:
 1. **A brief overview** of the whole computational framework of the thesis study, which is a 5-stage process with the analogy of 'matrix' to state what computations and operations are done in each stage.
 2. **A detailed description** for each stage with workflow figures. Each workflow figure showed the input and output values, computations and operations in that

stage, the way of validating the consistency of these computations and operations with Jian's study that was read from her thesis.

Oral explanation were given for specific steps in each stage/workflow. The way of validating the consistency of computations was using the full 311 original data as long as it could be done in that stage, then apply the same computations on the available 233 data.

- Prof. Javed suggested Feng with the following points:

1. On top of the brief overview of the whole computational framework, it needs the **detailed explanation** and concrete objects for each stage, like the data, models and its parameters.

Then, each stage/row in this overview slide should be augmented with 3 or 4 slides to show the details of the computations and operations in that stage.

Meanwhile, the findings (interpretations of the outputs) in each stage are also important, excessive technical details without attractive findings are not effective in a research presentation.

2. Feng should start **focusing on story telling now**, instead of developing more technical details. The story telling is about the questions, the methods to answer them, and the findings. It requires clarity and consistency in defining the targeted question, a good brief review of the related work, and the clear and understandable presentation of the methods and findings.

3. The completeness of the thesis is Feng's **first priority**, it should be accompanied with a good conceptual storyline to convince the readers.

The writing of the conceptual storyline should start at this moment, though some technical details and derivative results are not ready yet, but the main storyline should be clear to guide the writing.

4. The statistical and probability methods talked in this report are good, but they need to be connected to the ecological questions and serve for the story telling purpose in the thesis study.

- Prof. Jan gave feedbacks to Feng with the following points:

1. What is the **primary research question** that Feng is really trying to answer? Is this study trying to answer "How the benthic creatures response to the toxic chemicals in the study area, and how we extend this understanding?" or "How to design the statistical steps in a quantitative approach to explore the numerical relationship between the taxa composition (data) and the toxic chemicals (data)?"

It needs to clarify which one is the main question. It is reasonable that the planned work can solve both questions to some extent, but they should not carry equal weight or importance, one should be the main target and the other supplementary.

Such clarity in prioritization is essential for readability of the thesis, the coherence of the story telling, and finally the quality of the thesis and the expected articles from it.

2. There are two clear targets that Prof. Jan expects from Feng's work, they can be found in the quantile regression figures in Jian's thesis. In each figure/panel, the scatters are plotted and quantile regression lines are fitted with the two compressed representation vectors from taxa and chemical data/matrices respectively. The two targets are:

- (a) How to find and validate the **windows** (bottom left and upper right) in the scatter plot and the thresholds based on which the windows are derived. Specifically, the **bottom left** threshold and the origin define a rectangle window that shows an area in this panel where the reference sites can **not**

be found. The **upper right** threshold and the furthest point on the diagonal line shooting from the origin define a rectangle window that the degraded sites can **not** be found.

These two windows reveal a pattern in the relationship between the two latent representation vectors, one is for taxa composition and the other is for pollution level. The two blank windows tell us more certain information than does other area over the same interval of any of the two vectors, they define **exclusion regions** for potential new sites given such a panel structured from the data and intermediate computations.

- (b) Regardless of the observed pair of windows, we are interested in the **quantitative quantile relationship** between the two latent representation vectors, which enables the extension of the observed pattern in the scatter plot to the unobserved samples.

Notably (also an important attribution in this study), piecewise quantile regression model (PQRM) with breakpoint(s) is expected to be applied for the quantile regression. Then, the PQRM's sensitivity to sample sizes needs to be evaluated, that is how the confidence intervals of the PQRM change with different sample sizes and how well these CIs cover the 'true'/observed parameters of the model fitted with the full data set.

My reflections in connecting the CIs with the decisive windows

I propose referring to these regions as '**decisive windows**' because they are decisive in answering the 'yes or no' question about whether a site is reference or degraded.

The two windows are first derived from the observed scatter plot with nothing to do with the regression models, they are good (certain) in answering the 'yes or no' question about whether a site is **not** reference or degraded. But they do not tell us the 'how much' questions about the taxa composition and the pollution level of a site, and it is hard to explore the variation from such observed 'windows' to the unobserved samples. Nevertheless, this exploration into the variation of the windows can be complemented by **inferential analysis** about the unseen samples based on the fitted regression models. That is, the fitted regression models generate synthetic response values from which new windows can be derived, then the confidence intervals are constructed not only for the PQRM parameters but also for the decisive windows.

Following this idea, the way of **constructing synthetic response values** plays a critical role in determining the CIs of the PQRM, and then the CIs of the decisive windows.

Action Items

- Feng is going to prepare for the **details in each stage**, and present them **in a less technical way** with more focus on the story telling. Each row/stage in the overview slide should be explained with 3 or more slides to show what concrete data and models are used, and what findings are obtained from it.
- Feng will **start the writing work now**, even though some technical details and derivative results are not ready yet, but the main storyline is overall clear to guide the writing.

First of all, Feng will write a good abstraction for the **SSC** before the deadline of February 17, a good opportunity to start the story telling with a brief writing work.

- Feng will prioritize the findings in the quantile regression figures of Jian's thesis, which thread the intermediate results from early stage computations together and produce a clear, practical and attractive ending.

- Feng is going to set a schedule that plans the completeness of the thesis with **weekly work and goals**, the title, chapters and sections of the thesis are also needed to be decided without delay.
- The first meeting with Jian is pending to be decided, but it would not be too late and Feng will share some summary documents to Jian before the meeting.

Follow-Up

Temporary meeting time: February 19, 2:30 - 3:00PM (PDT)

Supportive materials

- Mid-term reports were written by supervisors and submitted to the program coordinator.